

WORLD FOOD PROGRAMME

Food Distribution Analysis — Kakuma Refugee Camp & Kalobeyei Integrated Settlement

Analysis Period: January – December (FY2023–2025 records) | Report Date: June 2026

1. Executive Summary

This report presents the findings of an analysis of 15,000 food distribution records covering refugee households across Kakuma Refugee Camp (Kakuma 1–4) and Kalobeyei Integrated Settlement (Villages 1–3). The objective was to assess distribution patterns, beneficiary coverage, operational performance, and equity of allocation across locations, nationalities, and distribution centers, to inform WFP management decisions on future food assistance operations.

A total of 15,000 households were served, reaching 97,491 beneficiaries and resulting in distribution of approximately 3.87 million kilograms of food across the reporting period. On average, each household received approximately 257.7 kg of food per cycle. Operations were supported by ten distribution centers (DC-01 to DC-10), distributed fairly evenly across the seven catchment locations.

15,000 Households Served	97,491 Total Beneficiaries	3,865,428 kg Total Food Distributed	257.7 kg Avg. Food / Household
10 Distribution Centers	33.9% Completion Rate	32.9% Delayed Rate	33.3% Partially Completed

Overall, distribution volumes are remarkably balanced across locations and nationalities, suggesting that allocation policies are being applied consistently. However, the analysis identifies a significant operational concern: only one-third of distributions were completed on schedule, with the remaining two-thirds either delayed or only partially completed, with average delays of 6.8–7.1 days across all seven locations. This indicates a systemic, camp-wide operational bottleneck rather than an issue isolated to specific sites.

2. Key Findings

2.1 Households and Beneficiaries Reached

The dataset records 15,000 household-level distribution events reaching a total of 97,491 individual beneficiaries. With an average household size that yields roughly 6.5 beneficiaries per recorded distribution, the program is shown to be reaching a broad cross-section of the camp and settlement populations. Food assistance was distributed almost equally between male- and female-headed households (Female: 7,522 distributions; Male: 7,478 distributions), reflecting balanced targeting with respect to the gender of the household head.

2.2 Distribution by Location

Food distribution volumes are highly consistent across all seven locations, indicating an equitable geographic allocation strategy. Kakuma 2 recorded the highest total distribution volume, while Kalobeyei Village 2 recorded the lowest, though the spread between the highest and lowest location is narrow (approximately 4.6%).

Location	Total Distributions
Kakuma 2	2,211
Kalobeyei Village 1	2,175
Kakuma 4	2,155
Kakuma 3	2,120
Kakuma 1	2,119
Kalobeyei Village 3	2,111
Kalobeyei Village 2	2,109

2.3 Distribution by Nationality

Beneficiaries from eight nationalities were served, with Ugandan refugees receiving the highest share of assistance and Ethiopian refugees the lowest. The variation across nationalities is moderate (roughly a 10% spread between the highest and lowest), suggesting that allocation is driven primarily by population size at each location rather than nationality-based disparities.

Nationality	Total Distributions
Uganda	1,975
DRC Congo	1,910
South Sudan	1,903
Burundi	1,886
Sudan	1,858
Rwanda	1,860
Somalia	1,813
Ethiopia	1,795

2.4 Distribution by Food Commodity

Six core commodities were distributed: Rice, Maize Flour, Beans, Salt, Super Cereal and Vegetable Oil. Distribution volumes are tightly clustered, with Super Cereal recorded most frequently and Salt least frequently — a difference of under 5%. This consistency suggests that the standard food basket composition is being applied uniformly across distribution cycles.

Food Commodity	Total Distributions
Super Cereal	2,561
Vegetable Oil	2,539
Rice	2,494
Beans	2,484
Maize Flour	2,470
Salt	2,452

2.5 Monthly Distribution Trends

Distribution volumes are spread fairly evenly across the calendar year, ranging from a low of 1,185 in June to a high of 1,326 in March. Modest peaks occur in March, May, August and October, while June, September and November show slightly lower activity. These fluctuations appear to follow

normal operational cycles rather than indicating any major seasonal disruption, though the recurring dip in June and around the September–November period may merit closer monitoring to confirm there is no underlying access or pipeline issue.

2.6 Distribution Center Performance

Ten distribution centers (DC-01 to DC-10) served beneficiaries fairly evenly, each handling between approximately 9,400 and 10,100 beneficiaries over the analysis period — a spread of roughly 6.6%. DC-03 served the largest number of beneficiaries (10,096), while DC-05 served the fewest (9,448). This narrow range indicates that center-level capacity is broadly matched to local demand, with no single center appearing significantly overburdened.

Distribution Center	Total Beneficiaries Served
DC-03	10,096
DC-06	9,943
DC-08	9,888
DC-02	9,806
DC-09	9,729
DC-07	9,712
DC-10	9,696
DC-01	9,648
DC-04	9,525
DC-05	9,448

2.7 Distribution Status and Delays

This is the most significant finding of the analysis. Of 15,000 distribution records, only 5,082 (33.9%) were marked Completed on schedule, with the remainder split between Delayed (4,929, 32.9%) and Partially Completed (4,989, 33.3%). Roughly two-thirds of all planned distributions experienced some form of disruption.

Distribution Status	Number of Records	Share of Total
Completed	5,082	33.9%
Delayed	4,929	32.9%
Partially Completed	4,989	33.3%

Average delay days are remarkably uniform across all seven locations (6.84–7.11 days). This uniformity in both delay rates and durations strongly suggests a systemic cause — likely a centralized supply pipeline, transport scheduling, or commodity stock arrival issue — rather than a problem specific to any single center or settlement.

2.8 Average Food Allocation per Household

The average quantity of food distributed per household is approximately 257.7 kg, consistent across all seven locations within a range of 257.46–257.86 kg. This confirms the food basket allocation formula is applied uniformly regardless of location, a strong indicator of equitable program design.

3. Recommendations

Based on the findings above, the following practical recommendations are proposed to strengthen food distribution operations across Kakuma and Kalobeyei:

1. Prioritize resolution of the distribution delay problem. With two-thirds of distributions either delayed or only partially completed, WFP should conduct a root-cause review of the supply chain feeding all ten distribution centers, focusing on commodity arrival schedules, transport contracts, and warehouse dispatch timing, since the uniformity of delays across locations points to a centralized rather than local cause.
2. Introduce real-time tracking of distribution status. A dashboard alert flagging distributions remaining "Delayed" or "Partially Completed" beyond 5 days would allow managers to intervene before delays compound.
3. Investigate seasonal dips in June, September and November. Although overall monthly volumes are stable, these recurring lower-activity months should be cross-checked against procurement and transport schedules to rule out early pipeline disruption.
4. Maintain the current equitable allocation model. Location, nationality, and per-household allocation volumes are highly consistent; this balanced approach should be preserved as a benchmark for any new centers or catchment adjustments.
5. Conduct targeted monitoring at DC-03 and DC-05. DC-03 serves the highest beneficiary load (10,096) and DC-05 the lowest (9,448); periodic visits should confirm DC-03 has adequate staffing and storage to sustain its higher throughput.
6. Strengthen data quality controls at the point of distribution, including validation rules for date formats, standardized nationality naming, and quantity range checks, to ease future analysis and improve reporting reliability.
7. Use the interactive dashboard for monthly management review, tracking whether the completion rate (currently 33.9%) improves in response to corrective actions.

4. Conclusion

The analysis of 15,000 food distribution records across Kakuma Refugee Camp and Kalobeyei Integrated Settlement demonstrates that WFP's food assistance program is reaching a large and diverse population — 97,491 beneficiaries — through a well-balanced distribution model. Allocation of food commodities is consistent across the seven locations, eight nationalities, and six food commodity types, with an average household allocation of 257.7 kg that varies by less than half a kilogram between locations. Beneficiary loads across the ten distribution centers are similarly well-matched to demand, and gender representation among household heads is nearly equal.

The single most pressing operational issue identified is timeliness: only one in three distributions (33.9%) were completed as scheduled, while the remainder were delayed or only partially completed, with average delays of roughly 7 days that are consistent across every location. Because this pattern is uniform camp-wide, it points to a systemic supply chain or scheduling issue rather than localized inefficiency, and should be the top priority for WFP management going forward. Addressing this challenge — while preserving the program's existing equitable allocation framework — represents the clearest opportunity to improve outcomes for refugee households in Kakuma and Kalobeyei. The cleaned dataset, pivot tables, and interactive dashboard accompanying this report provide WFP management with the tools needed to monitor progress against these recommendations on an ongoing basis.